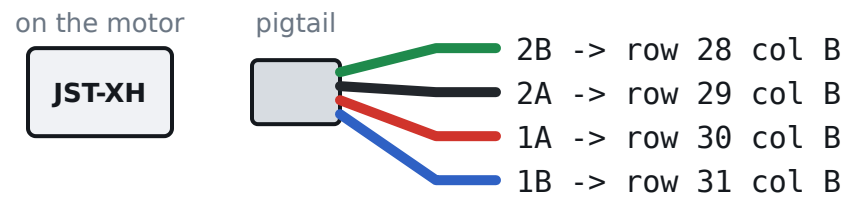


NOTHING HERE NEEDS ACQUIRING. Every item was confirmed on hand on 2026-09-11, the 830-point breadboard included. This box exists so that anything a later revision adds lands on page 1 rather than being discovered on page 13.

☐ TMC2209 StepStick (FYSETC) 1 of 6 The driver. Headers arrived PRE-SOLDERED, confirmed by opening the box, so page 5 is a skip. All six are one revision, so one Vref carries to all five later.	☐ 100 uF electrolytic, 35 V or more 1 The bulk capacitor. Its legs need trimming and bending before it will reach the right two holes - lead prep on page 2.	☐ Hex key for the drum set screw 1 Fit the stand-in drum, and MARK it - a tape flag will do - so that `step 0 3200` is a readable result rather than a guess.
☐ ESP32-C5-DevKitC-1 1 Already on the bench. It stays OFF the breadboard - page 8 says why - so every wire that reaches it is male-to-female.	☐ Hookup wire, 22 AWG SOLID core 30 cm Red and black, PSU terminal to the board. Solid, not stranded: stranded frays in a breadboard hole, will not hold, and leave a strand behind.	☐ Heatsink (supplied, loose) not used Deliberately not installed. Page 2 has the reasoning and the one condition that would change it.
☐ NEMA 17 + printed stand-in axle 1 The load. The axle is PLA, which is why every commanded speed is REFUSED above this image's cap. The cap is compiled in and can only be lowered at build time, never raised and never from a console.	☐ Flat screwdriver, 2 mm blade 1 The Vref trimpot on page 27, and the PSU's terminals. A blade that does not fit the pot slips off it onto a pad.	☐ A1121LUA-T Hall sensor 1 NEW THIS SESSION - gate 3 had none. 3-pin SIP, not TO-92. Page 23 says which way round it goes; the centre lead is GND whichever way you read it.
☐ Breadboard, full-size 830-point 1 63 rows, two rail pairs, an 0.3 in centre channel. The channel is the only reason a StepStick works on a breadboard at all - page 8.	☐ Multimeter with a continuity beep 1 Coil pairs (page 9), continuity (page 26), Vref (page 27). Continuity and DC volts on a 2 V range are all it needs.	☐ Magnet, O6x3 N42, in the disc fitted You are not prising it out to look at it. Page 25 reads its outward face with a SPARE magnet instead: repel = south = correct.
☐ Jumper wires, male-to-FEMALE 8 Three signals (STEP, DIR, EN), the two rail feeds, and the Hall's 3 flying leads. Female onto the ESP32's pins, male into the board.	☐ USB-C cable, PC to ESP32 1 Console and logic power. A DATA cable - a charge-only lead gives a board that looks alive and never appears as a serial port.	☐ Spare magnet, same batch 1 The polarity meter. It identifies its own south face against the sensor, and then reads the glued one without touching it. Page 25.
☐ Jumper wires, male-to-male 6 Board-internal: VIO, GND (logic), MS1 and MS2 to the rails, plus 2 RAIL LINKS: the two rail pairs are separate nodes and both are in use. Counts derive from the parsed table.	☐ Bench PSU, current-limited 1 VM, and the most useful safety device on the bench. 9 V at a 0.5 A LIMIT for first power, 20 V for the soak. A wiring fault trips the limit, not the driver.	☐ Resistor, 10 kohm, any type 1 The Hall's pull-up, OUT to 3V3. The output is OPEN DRAIN: it can pull the pin down and cannot pull it up, so without this `hall` reads noise.
☐ JST-XH 4-way mating pigtail 1 THE MOTOR PLUG CANNOT ENTER A BREADBOARD. The pigtail's four flying leads can. Page 2 shows exactly where they terminate.	☐ RotoPD USB-C trigger -- NOT USED struck I2C-configured and DEFAULTS TO 5 V, so it cannot be trusted to come up at the voltage you set, and it current-limits nothing. Struck, not deleted, so it is not bought again.	☐ Ceramic capacitor, 0.1 uF 1 Across the sensor's VCC and GND, at the sensor end. The datasheet states it in words and the flying leads are long.

Three parts need preparing, one decision is already made, and one thing is deliberately NOT checked

1 The motor plug cannot enter a breadboard



The loom ends in a 4-way JST-XH. Its shell will not go into an 0.1 in board and forcing it spreads the contacts permanently. The mating pigtail plugs onto it and ends in four bare leads: strip 6 mm, twist each hard or tin it, and push them into rows 28 to 31 of column B. WHICH lead goes where is decided by the meter on page 9, never by colour - these wires were cut and re-terminated.

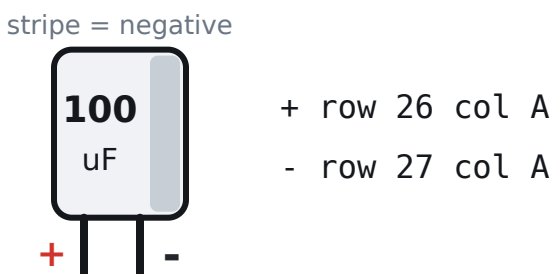
2 The bench supply reaches the board on two wires



Strip 8 mm of the 22 AWG solid core, clamp it in the terminal, and run the other end straight into the driver's own row. VM never goes through a rail.

SET THE CURRENT LIMIT BEFORE THE VOLTAGE. 9 V at 0.5 A for first power: the whole column draws under 0.1 A at 0.7 A RMS coil current, so anything near the limit is a fault and the supply folds back instead of the driver dying.

3 The capacitor needs its legs prepared



It bridges the driver's OWN VM and GND, which are adjacent rows - 2.54 mm apart, against the 3.5 or 5 mm the leads came at. Trim both to about 12 mm, bend them toward each other at the body, and check the bends cannot touch. The body may overhang other rows; only the legs matter. Backwards, it vents.

4 Heatsink: not needed this session

DECIDED: the driver runs bare. 0.7 A RMS against the module's 2 A rating is roughly an eighth of rated dissipation, and the soak's thermal question is the motor sealed in PLA, not the driver. Page 32's STOP NOW list already carries 'the driver too hot to keep a finger on', so this is watched rather than assumed.

Not independently verified: the exact RDS(on)-derived junction temperature. IF one ever goes on, the adhesive pad MUST clear the Vref trimpot - a heatsink overhanging the pot ends page 27.

5 The current path, so nobody over-engineers

0.7 A RMS through 22 AWG wire, stock jumpers and breadboard rails is fine and needs no thought. A breadboard contact is good for about 1 A continuous and a jumper drops tens of millivolts here. Two things are worth KNOWING rather than fixing: the 0.7 A coil current flows in the four motor leads and inside the driver, never along a rail; and VM and GND come in on their own wires to the driver's own rows to keep the switching return out of the logic ground, not for current rating. The bulk capacitor, not the supply wire, delivers each step's transient. Upgrade nothing.

6 What the guard checks, and what it cannot

Every PIN on every page is parsed out of hal/pins.h and docs/BENCH_WIRING.md, cross-checked against the other, and the generator emits nothing at all if they disagree. That is why a drawing here cannot quietly contradict the firmware.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

It is the same distinction page 6 draws between a LABEL and a drawn position. The coordinates are chosen in tools/wiringgen.py, they are consistent with each other, and consistent is not verified.

The whole job in 32 pages, in the order you should do it. One connection per page. Every pin number here is read out of hal/pins.h and docs/BENCH_WIRING.md when these pages are rendered, so a drawing cannot quietly disagree with the firmware. Row and column numbers are a different kind of statement - page 2 says which.

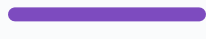
THE HALL SENSOR AND MAGNET ARE FITTED (2026-09-20). This module homes, so pages 23-25 are new and everything after them can be closed loop. Gate 3 ran without them and every page said so.

READ PAGE 4 FIRST. The three rules on it are how TMC2209 drivers die.

THE ORDER, and why it is this order

- 1-2 collect the parts, and prepare three of them
- 3-5 the rules, and the one solder job (skipped)
- 6-9 identify the parts, the board, the coil pairs
- 10-14 wire the logic (driver to ESP32)
- 15-19 wire the microsteps, then the power
- 20-22 the bulk capacitor, then the board as BUILT
- 23-25 the Hall sensor <- and PROVE its polarity
- 26 continuity check <- find mistakes with a meter
- 27 set Vref <- motor still NOT connected
- 28-29 connect the motor <- power off while you do it
- 30-32 finished layout, the built board, power on / off

WIRE COLOURS — the same on every page

	VM (motor +)	RED, thick
	GND (power)	BLACK, thick
	VI0 -> 3V3	red, thin
	GND (logic)	black, thin
	STEP	orange
	DIR	yellow
	EN	white / grey
	MS1	violet
	MS2	grey-blue
	PDN	brown
	HALL OUT	green, thin

The motor keeps its own colours: red, blue, green, black.
Use any wire you like — but be consistent, and **never use red or black for a signal.**

1

NEVER connect or disconnect the motor while VM is live.

An open coil on a live driver dumps its stored energy into the output stage. This kills more StepSticks than every other cause put together. Power down FIRST, then touch the motor plug — every time, including 'just for a second'.

2

NEVER insert or remove the driver module while VM is live.

Same physics, and you will bridge something on the way past as well.

3

Set Vref BEFORE the motor is ever connected.

Vref sets the coil current. Set it with the motor attached and the first thing your motor sees is wherever the pot happened to be when it left the factory — often well over 1.5 A. This guide sets Vref on page 17 and connects the motor on pages 28-29, in that order, deliberately.

And one that is less dramatic but more common

The bulk capacitor must sit at the driver's OWN VM and GND pins — not on a breadboard rail at the far end of the board. Page 20 shows where.

If you are unsure at any point: power off, and nothing you do next can break anything. Every irreversible mistake in this guide needs VM to be live.

SKIP THIS PAGE AND GO TO PAGE 6. The FYSETC modules arrived PRE-SOLDERED - confirmed by opening the box, not inferred from a product photo. The page stays because the next module to arrive may not be, and because the orientation below is the one thing that cannot be undone.

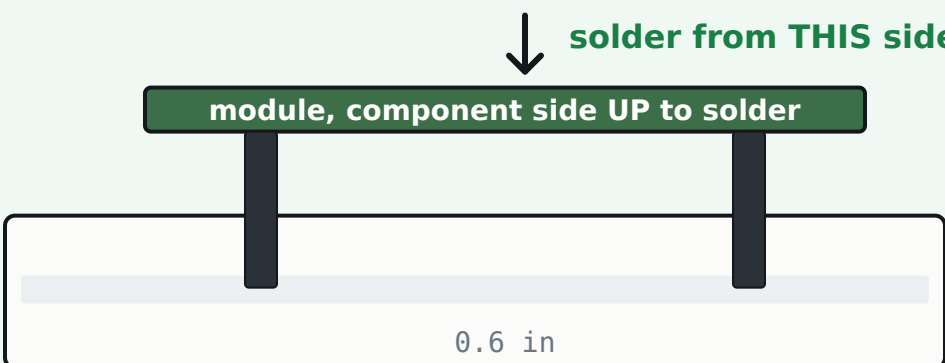
Confirm it in five seconds before you move on

- ☐ Two black strips of 8 pins each are IN the module, not loose in the bag.
- ☐ Every pin has a shiny cone of solder where it meets the board.
- ☐ Sighted from the end, all sixteen pins are parallel and the same length.
- ☐ The pins exit the COMPONENT face - the same face as the Vref pot. That is how these arrived, and page 7 says what follows from it.

If a future module arrives with loose strips

Solder it STRADDLED INTO THE BREADBOARD. The board holds all sixteen pins square and at the right spacing while you work, which is the whole problem - a module soldered freehand comes out with splayed pins that will not enter a breadboard, and straightening them afterwards breaks them at the joint.

RIGHT - pins held by the board



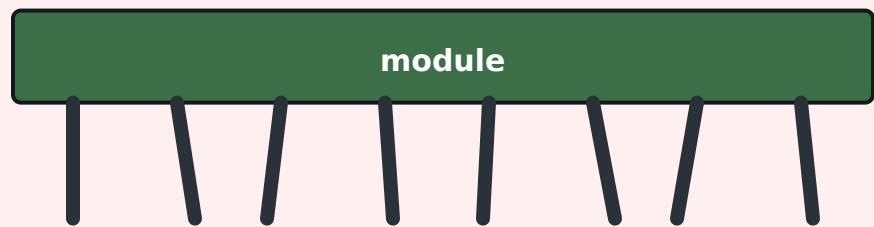
module, component side UP to solder

solder from THIS side

0.6 in

Strips into the board pins-down, module dropped on top, solder the sixteen joints you can see. MATCH THE FACE THE EXISTING ONES USE - page 7.

WRONG - pins held by hand or tape



module

splayed - will not enter a board

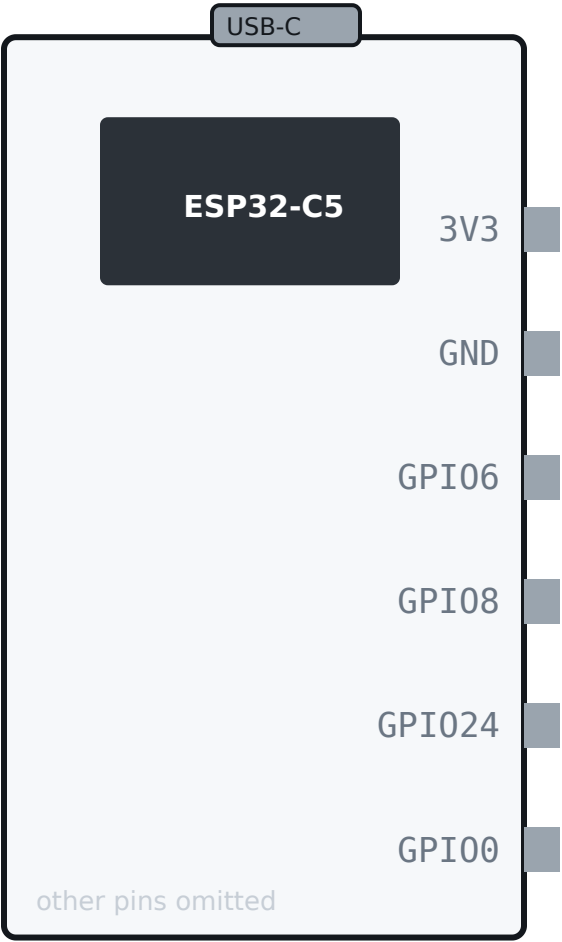
Sixteen pins cannot be held square by hand. This is the failure that costs a module rather than a minute.

Identify your two parts

Match by LABEL — drawn positions are only a hint

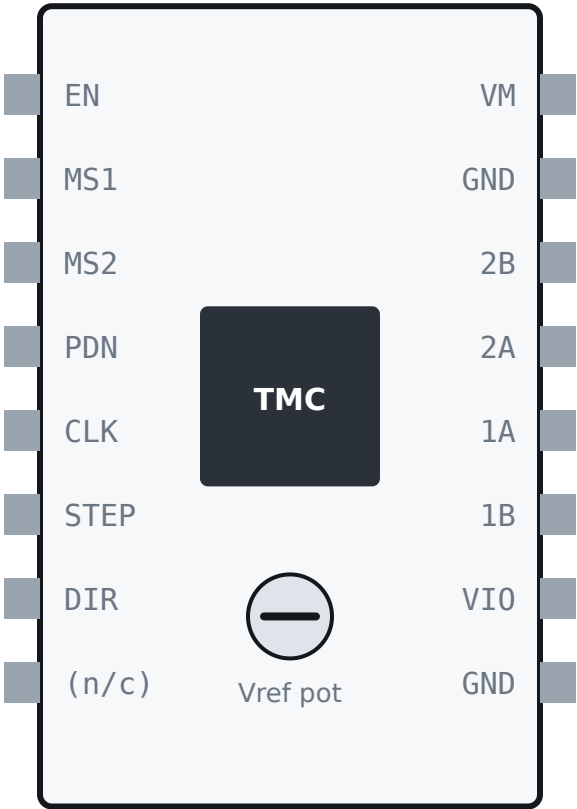
ESP32-C5-DevKitC-1

USB-C at the top



FYSETC TMC2209

standalone · match by LABEL



Pin ORDER here is the standard StepStick arrangement. If your module's silkscreen puts a label somewhere else, THE LABEL WINS. Every instruction in this guide names a pin and never a position.

Tick off these labels on the driver:

- | | |
|------------------------------------|-------------------------------------|
| ☐ EN | ☐ MS1 |
| ☐ MS2 | ☐ PDN / UART |
| ☐ STEP | ☐ DIR |
| ☐ VM / VMOT | ☐ GND (x2) |
| ☐ 1A | ☐ 1B |
| ☐ 2A | ☐ 2B |
| ☐ VIO / VDD | |

Find the two SENSE RESISTORS as well

Two small surface-mount parts beside the motor output pins, marked R110, R150 or R100. That marking decides your Vref target on page 27, and reading it takes ten seconds where guessing it costs you the whole thermal result.

R _____ = 0. _____ ohm

R110 = 0.11 ohm · R150 = 0.15 ohm · R100 = 0.10 ohm

Measured on the real part - and it is not the way a StepStick diagram draws it

THE HEADER PINS EXIT THE COMPONENT FACE - the face with the Vref trimpot on it. So the module seats POT-DOWN, against the breadboard. Two things follow, and the second one is the trap.

1. The pot is unreachable once the module is in

It is pressed flat against the board. There is no setting it, checking it or correcting it without pulling the module back out - and pulling a module out of a live board is how they die. So VREF IS SET BEFORE INSERTION, with the driver lying on the bench on two flying leads. Page 27 is that procedure, and it is why that page comes before the module ever goes in.



pins and pot on the SAME face
so the pot faces the board

2. Left and right swap. This is the trap.

The silkscreen names its two headers as you read them component-side UP. Turning the module over exchanges them. The order ALONG each header does not change - EN is still at the same end - so every pin is exactly where you expect along the module and on the wrong side of it. A mirrored VM and GND is a reversed supply, and nothing on the board will tell you.

CHECK IT WITH THE METER, NOT WITH YOUR MEMORY: continuity from the driver's VM pin to row 26 column D. That single beep settles which way round it went in.

Where each header actually lands

the silkscreen's LEFT header		column H
EN	row 26	col H
MS1	row 27	col H
MS2	row 28	col H
PDN	row 29	col H
CLK	row 30	col H
STEP	row 31	col H
DIR	row 32	col H
(n/c)	row 33	col H

the silkscreen's RIGHT header		column D
VM	row 26	col D
GND	row 27	col D
2B	row 28	col D
2A	row 29	col D
1A	row 30	col D
1B	row 31	col D
VI0	row 32	col D
GND	row 33	col D

3. The aux pins: BEND, do not cut

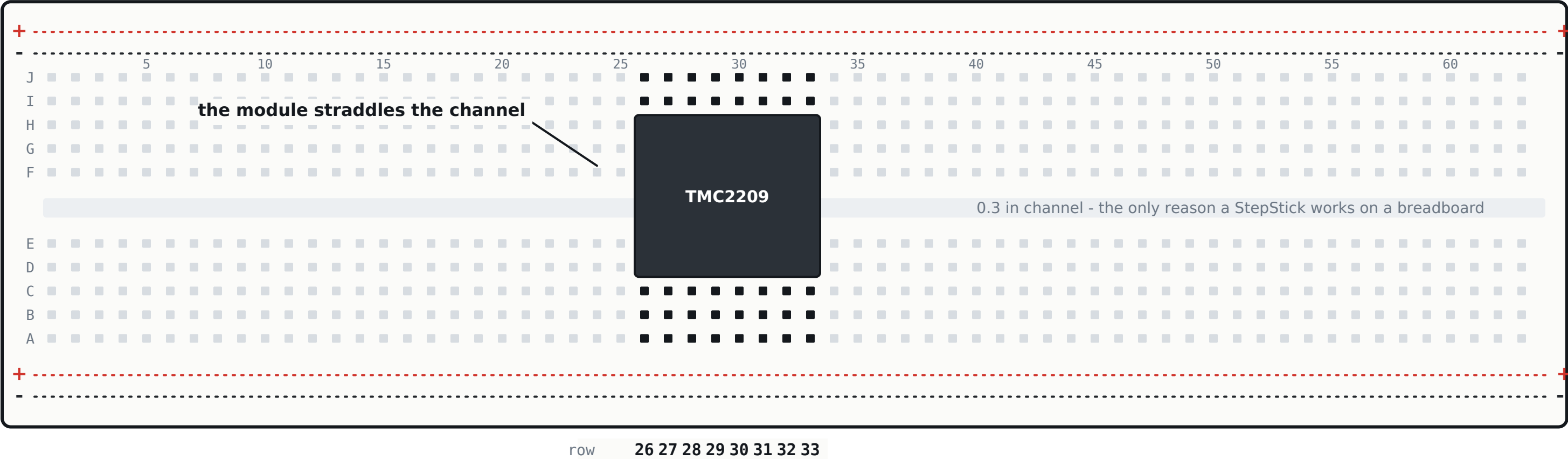
The pins this build does not use - PDN/UART, CLK, and the second GND - still have to go somewhere. Bend them flat against the module body before you seat it. Do NOT cut them: the module is one of six, the other five may want UART later (spec 5.7a plan B), and a cut pin cannot be put back. A bent pin straightens with pliers in ten seconds.

A bent pin must not touch its neighbour or the board. Check it before it goes in - once it is in, you cannot see it.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

Nothing is wired yet. This is the board, drawn as it is.

Two facts about this board do all the work. Each group of five holes across ONE row, on one side of the channel, is ONE electrical node - so a wire in row 26 column A is electrically the driver's VM pin. And the four long strips along the edges are the POWER RAILS, which are joined to nothing at all until page 21 makes them live.



Why the driver lands in columns H and D

A StepStick's two headers are 0.6 in apart. Counting across the channel, column H to column D is $0.1+0.1+0.1+0.3+0.1+0.1 = 0.6$ in exactly, so the module drops in with no pin sharing a node with its opposite number. That spacing is the one certain physical fact these coordinates are built on.

COLUMNS E, F AND G DO NOT EXIST for this build: the module body sits over them. Every wire on the right half goes into I or A, every wire on the left half into J. That is clearance, not tidiness.

THE ESP32 STAYS OFF THE BOARD

Its header-to-header spacing is 0.9 in or 1.0 in depending on the inset, and on an 0.1 in board with an 0.3 in channel both of them fit - landing the pins in different columns. Which GPIO sits where along the header is a board fact this repository does not carry, and guessing bends pins. So the DevKitC-1 lies loose on the bench and every wire to it is male-to-FEMALE.

Find these five by label on the silkscreen:

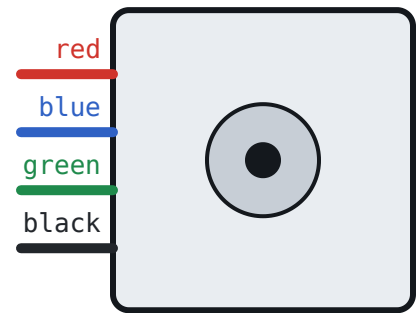
- ☐ GPIO8 (STEP)
- ☐ GPIO24 (DIR)
- ☐ GPIO6 (EN)
- ☐ 3V3
- ☐ GND

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

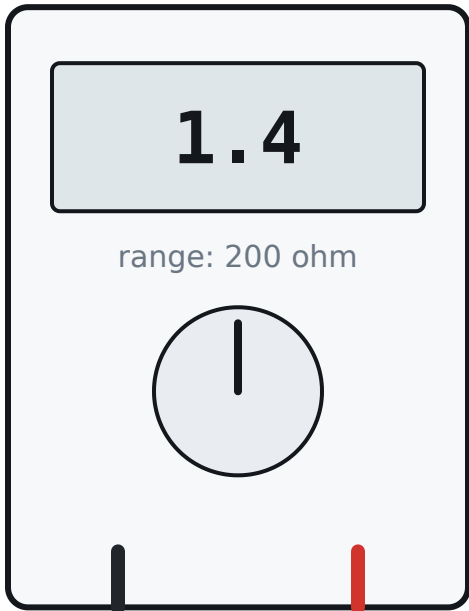
Find the coil pairs

Motor connected to NOTHING. Meter only.

LDO 42STH48-2504AH



DMM



The wires were cut and re-terminated, so no colour convention is evidence any more. Measure all six combinations. Exactly two read low — those are the coils. The other four must read open.

FIRST touch the probes together and note the reading (0.1-0.5 ohm). That is your test leads. Subtract it from everything below.

red - blue	_____	ohm
red - green	_____	ohm
red - black	_____	ohm
blue - green	_____	ohm
blue - black	_____	ohm
green - black	_____	ohm

CROSS-CHECK WITH NO METER: twist one candidate pair's bare ends together and turn the shaft. A real pair makes it noticeably harder and notchier to turn; a wrong pair turns freely. Ten seconds, and it catches mistakes a meter reading does not.

Write it down — pages 28 and 29 refer to it:

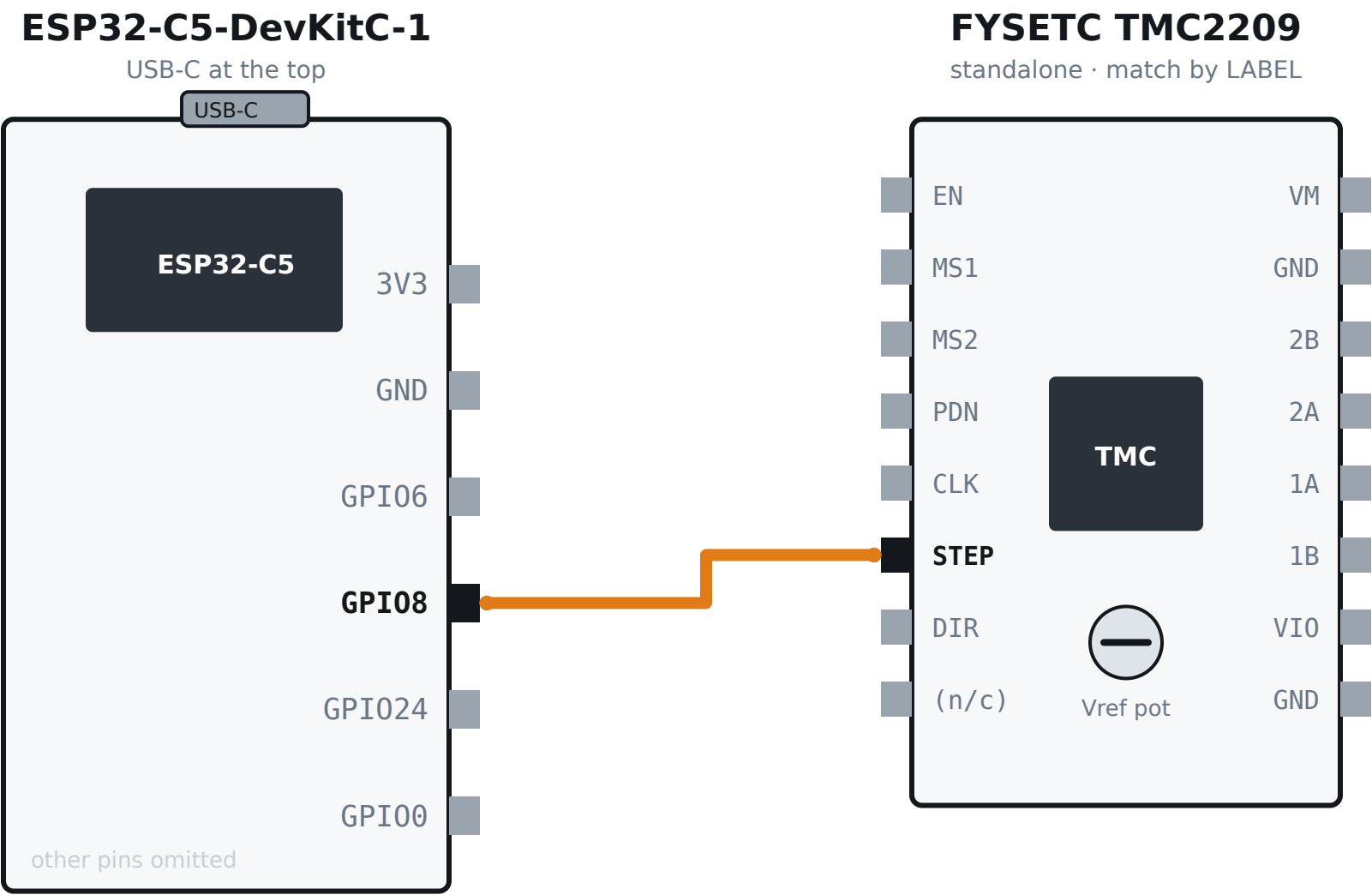
PAIR 1 = _____

PAIR 2 = _____

Expected for this motor: black+green one coil, red+blue the other. IF YOUR METER DISAGREES, THE METER WINS.

Two low (about 1-2 ohm), four open. More than two low is a short — usually a stray strand on a Dupont pin. Any wire beeping to the motor CASE: stop, that winding is shorted to the frame.

wire colour: orange

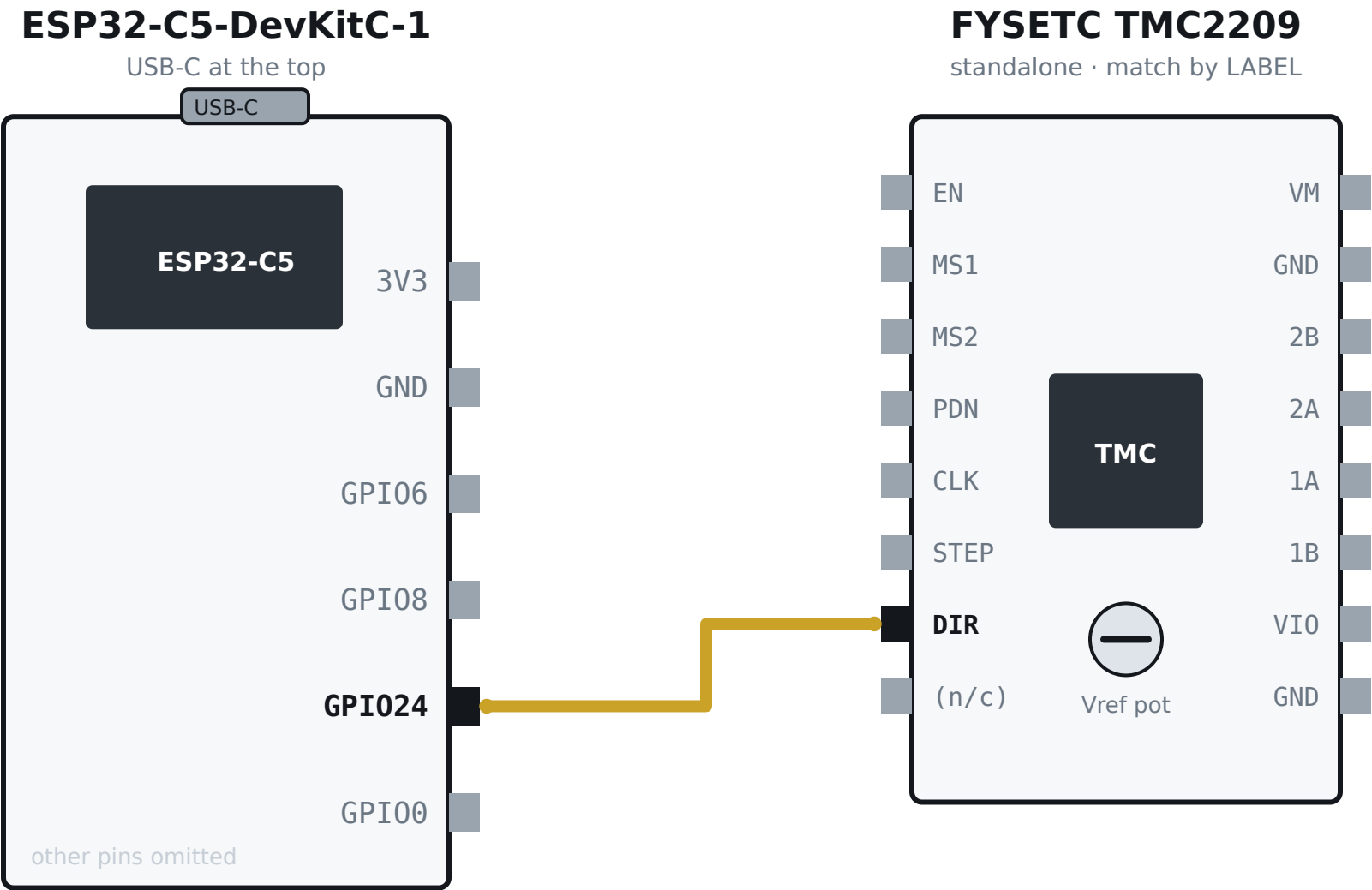


One pulse on this wire moves the motor one microstep. At 1/16 microstepping that is 1/64th of a flap, and 3200 pulses is exactly one drum revolution. This is the only wire that carries motion.

ON THE BOARD

STEP is row 31 column H. Jumper (M-F) row 31 column J -> the ESP pin labelled GPIO8.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

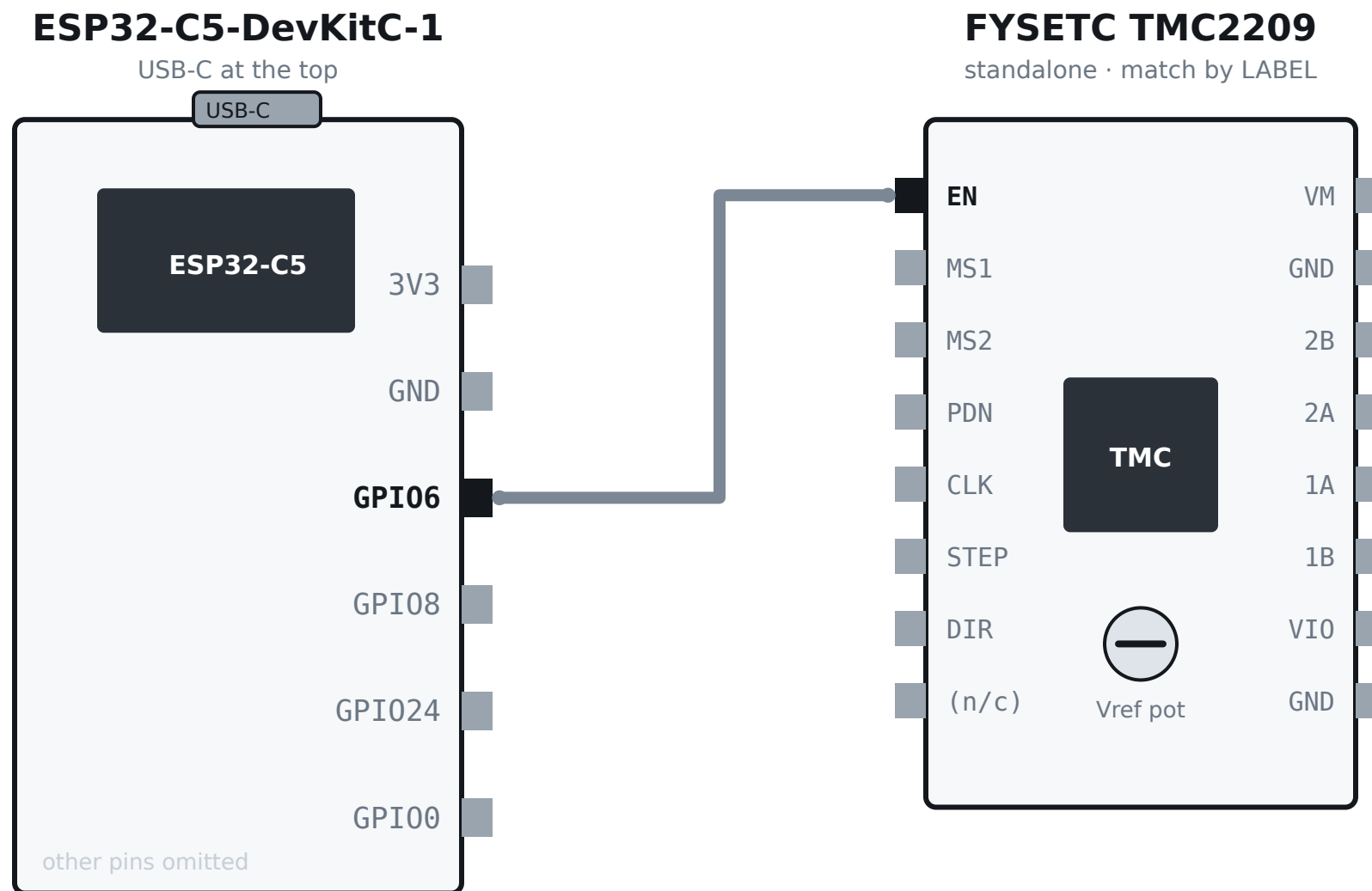


A level, not a pulse: the driver reads it on each STEP edge to decide which way to turn. The ring is descending, so getting this backwards makes a countdown count up — which is why it is a firmware bit you can flip with `dir` rather than something to get right with a soldering iron.

ON THE BOARD

DIR is row 32 column H. Jumper (M-F) row 32 column J -> the ESP pin labelled GPIO24.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.



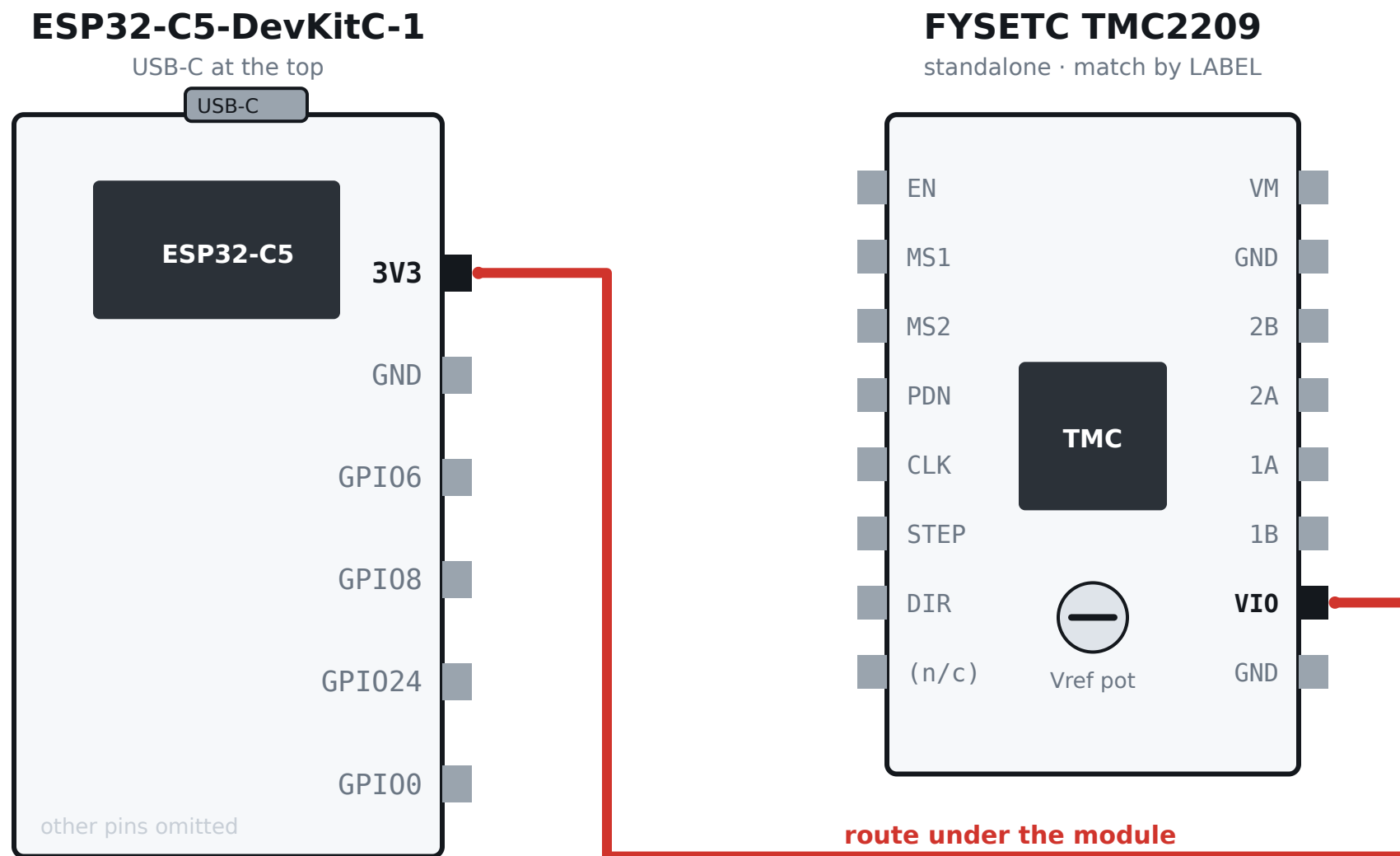
ENABLE, and it is ACTIVE LOW: pulling it low turns the coils on. Most modules pull it high on board, so an unconnected EN means disabled — which is the safe default you want for a first power-on.

CHECK YOURS. With USB in and VM off, measure EN against GND: it should read about 3.3 V. If it reads near 0 V, add a 10 kohm resistor from EN to 3V3 before you ever connect VM, or the coils energise the instant power arrives.

ON THE BOARD

EN is row 26 column H. Jumper (M-F) row 26 column J -> the ESP pin labelled GPIO6.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.



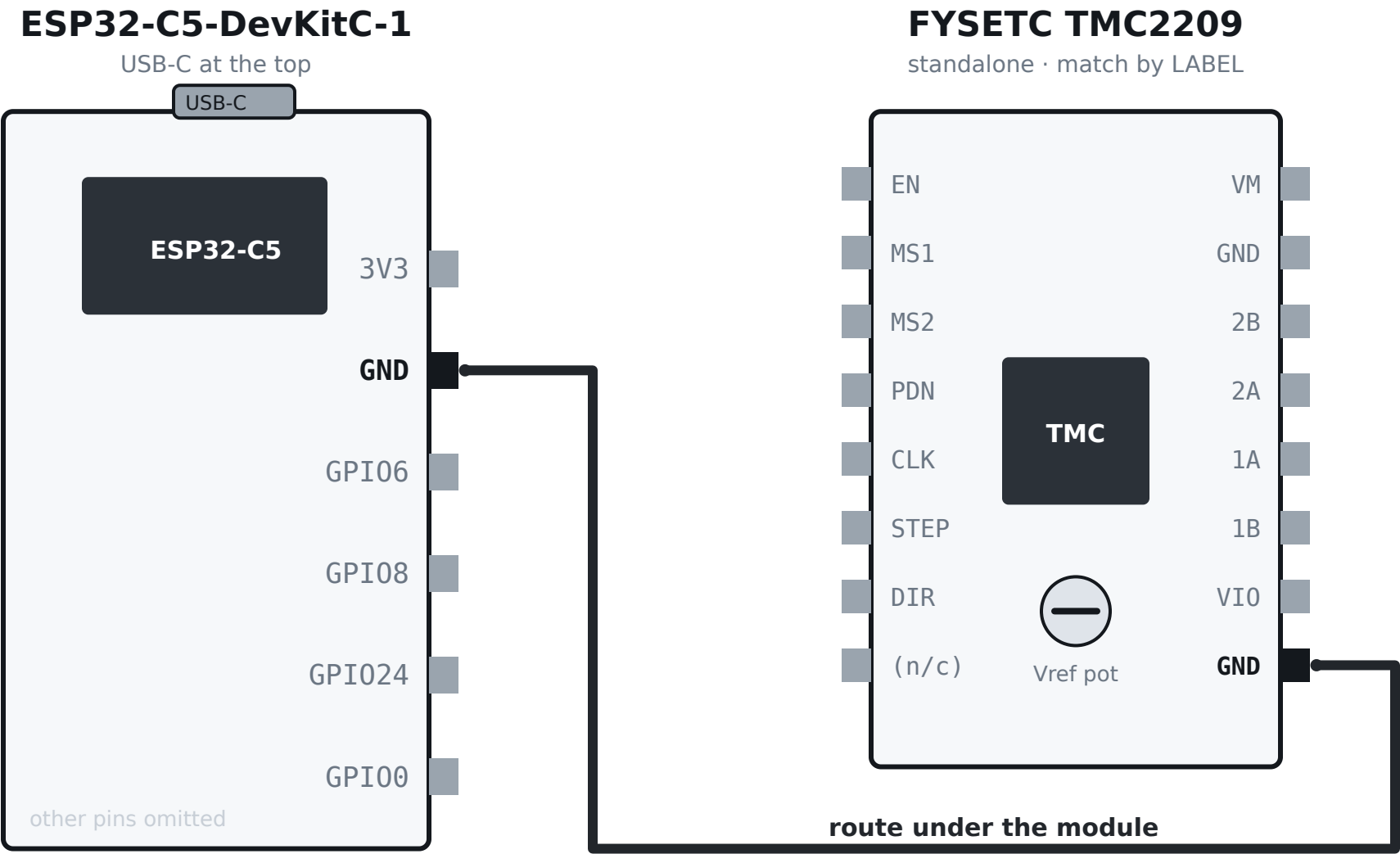
The logic supply. It sets the voltage the driver expects to see on STEP, DIR, EN and the MS pins, so it must be the SAME 3.3 V the ESP32 drives with. VIO is on the right-hand pin column, so this one routes under the module.

3.3 V, not 5 V. Feeding VIO 5 V makes the driver want 5 V logic levels from a 3.3 V microcontroller. It appears to work, right up until it does not.

ON THE BOARD

VIO is row 32 column D. Jumper (M-M) row 32 column B -> the + rail.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

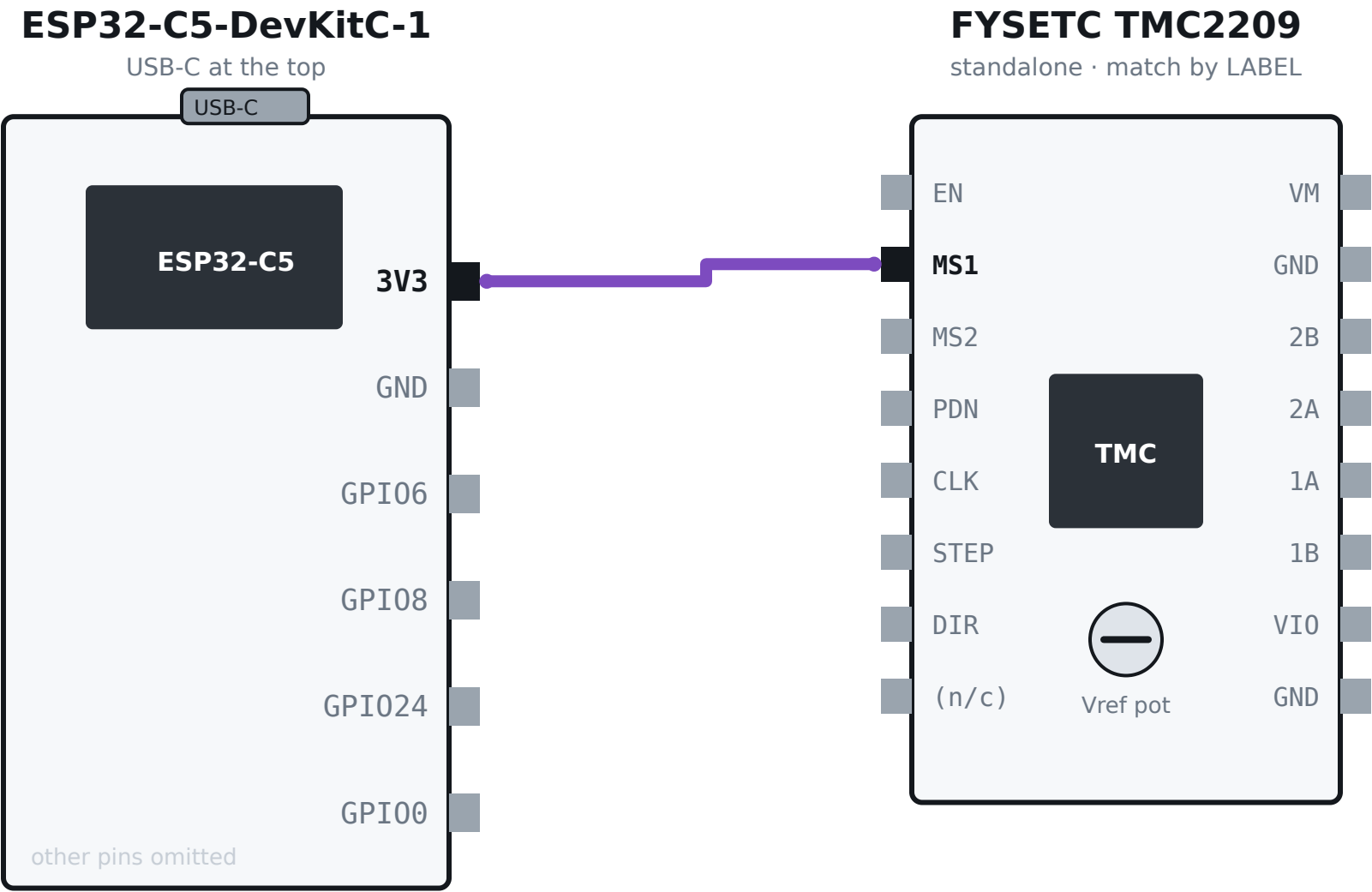


The logic ground. Without a shared ground the ESP32 and the driver have no common reference and every signal level is meaningless. This is the wire whose absence produces the most baffling symptoms.

ON THE BOARD

GND is row 33 column D. Jumper (M-M) row 33 column B -> the - rail.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.



Microstep select, bit one. Tie it HIGH.

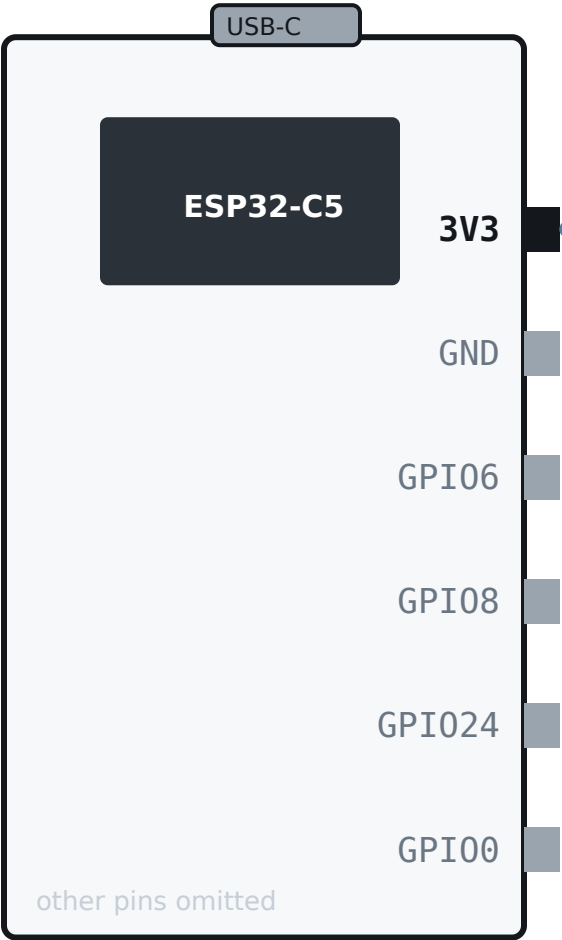
ON THE BOARD

MS1 is row 27 column H. Jumper (M-M) row 27 column J -> the + rail.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

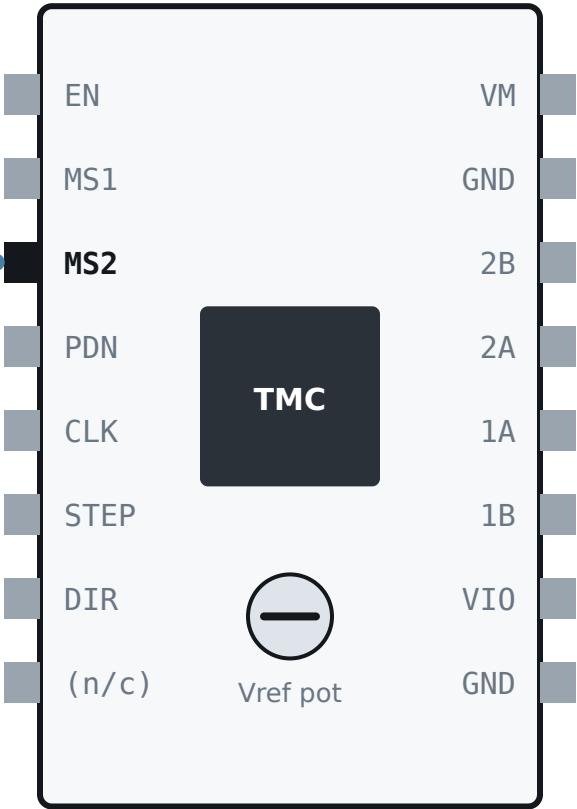
ESP32-C5-DevKitC-1

USB-C at the top



FYSETC TMC2209

standalone · match by LABEL



TMC2209 microstep table

MS2	MS1	microsteps
0	0	1/8
0	1	1/2
1	0	1/4
1	1	1/16

<- what we want

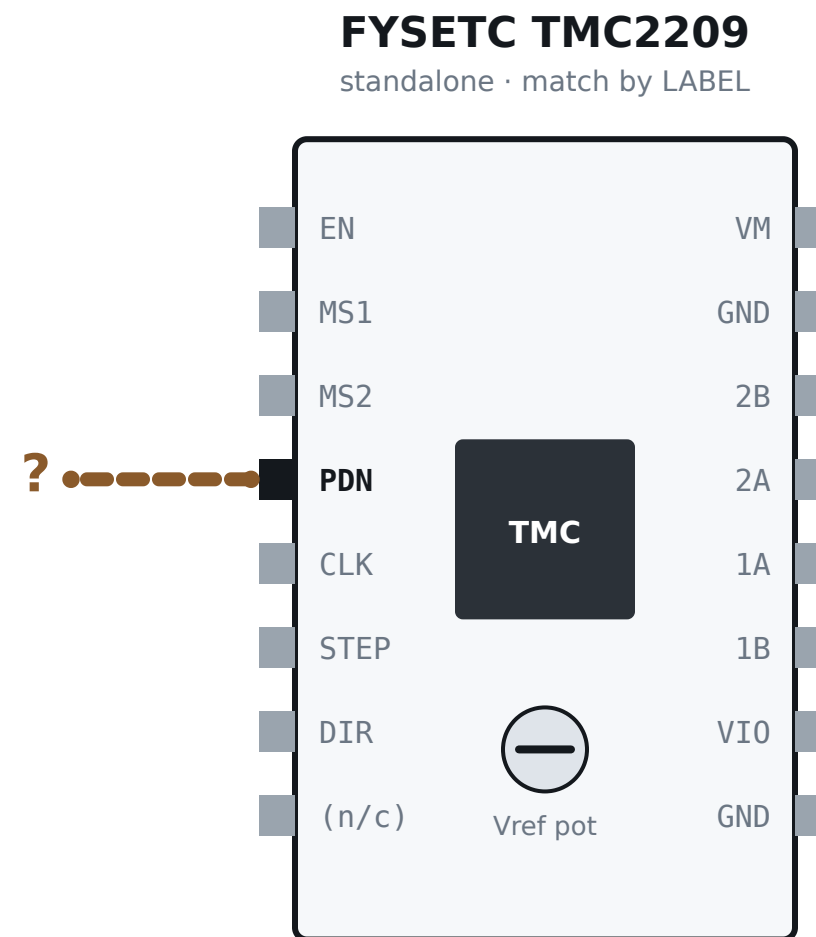
Microstep select, bit two. Tie it HIGH as well. BOTH high is 1/16 on a TMC2209, and the table is not in a sensible order — which is exactly why this one is easy to get wrong.

Modules usually pull MS1 and MS2 LOW on board. Leave them unconnected and you get 1/8, which makes every constant in the firmware wrong by a factor of two. Page 30 has a mechanical check that proves the setting — do not trust the jumper.

ON THE BOARD

MS2 is row 28 column H. Jumper (M-M) row 28 column J -> the + rail.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.



LEAVE AS THE MODULE SHIPS

for standalone operation

then VERIFY on page 32

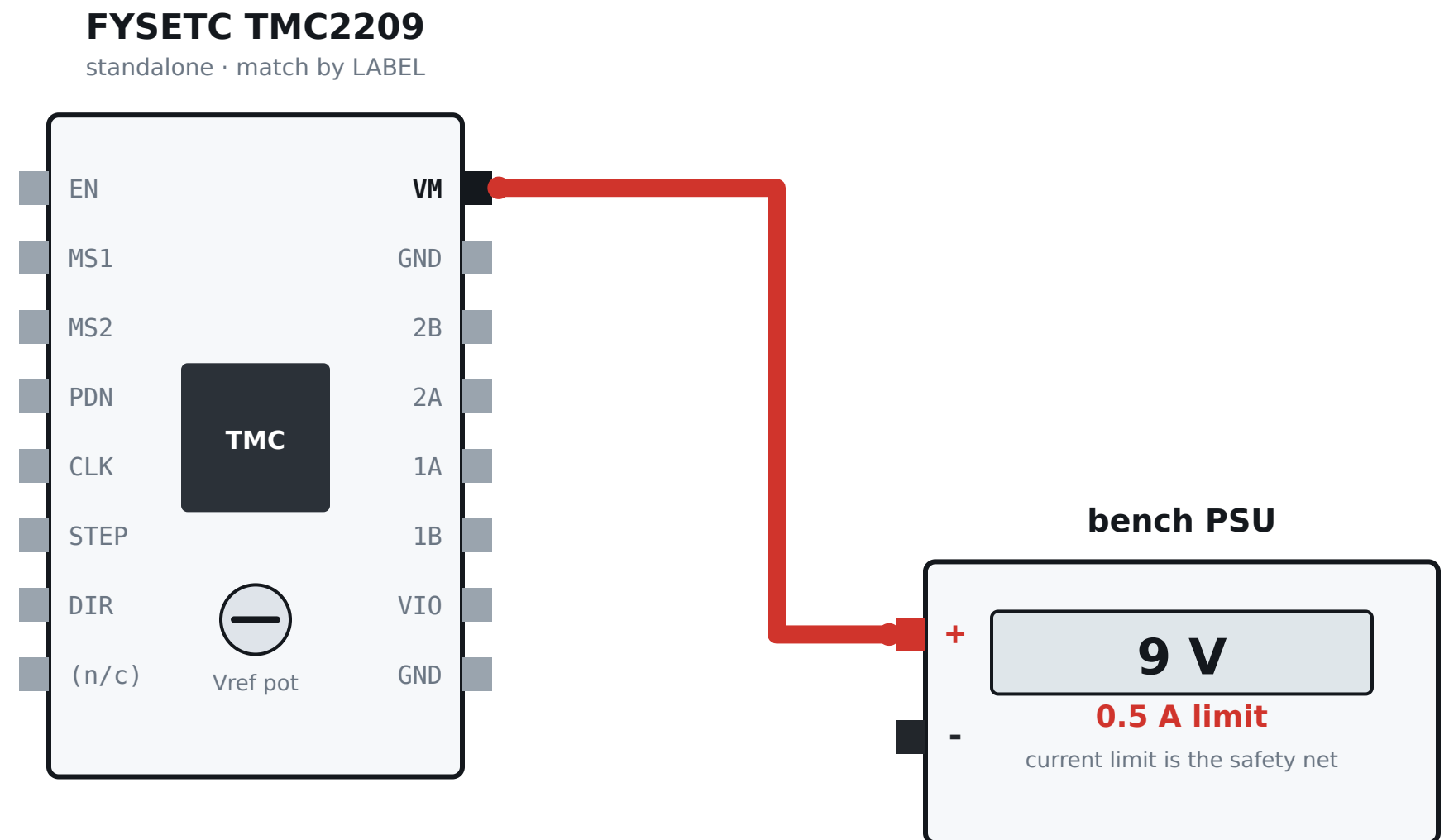
In standalone mode this pin controls automatic standstill current reduction: about a second after the last step, the driver drops the coil current to roughly half. That is not a nicety here — it is the entire thermal contract, because the motor is sealed in a PLA drum and holds current all day.

THE POLARITY IS NOT VERIFIED for your module and this guide will not guess it. Leave PDN as the module ships for standalone, then confirm on page 32: with the motor energised and still, the supply current must visibly STEP DOWN about a second after the last step. If it never does, tie the pin the other way and re-check.

ON THE BOARD

PDN is row 29 column H. NOTHING goes in row 29. Leave the whole row empty.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

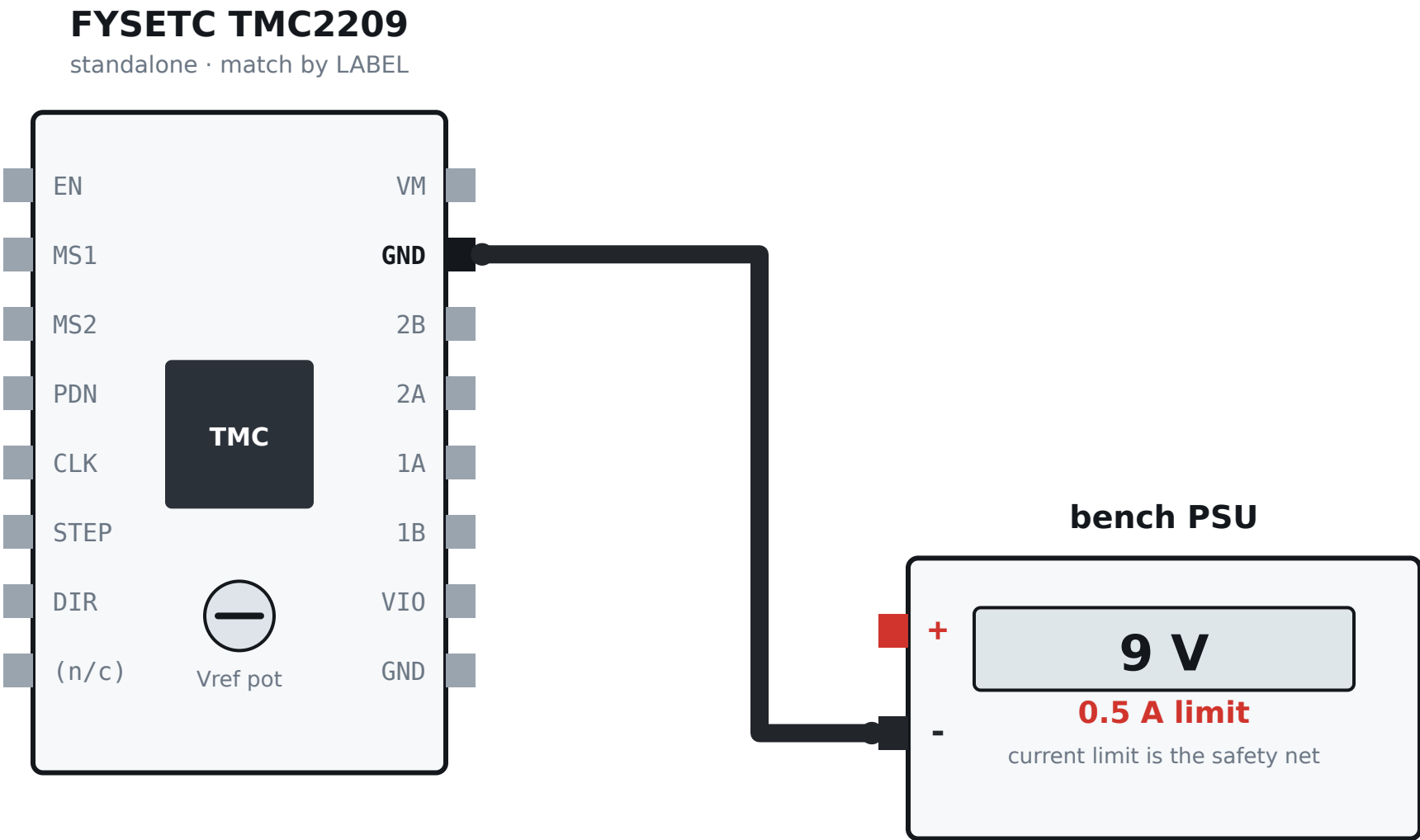


The motor supply, from a CURRENT-LIMITED bench PSU. Start at 9 V, not 20 V: the TMC2209 runs from 4.75 V up, at one drum revolution per second there is no headroom needed, and a wiring mistake at 9 V dissipates about a fifth of the energy. Move to 20 V once the wiring is proven.

ON THE BOARD

VM is row 26 column D. Bench PSU + on its own 22 AWG wire into row 26 column C.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from `hal/pins.h` and `BENCH_WIRING.md` and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.



The power ground — the wire that carries the coil current back. Keep it short and direct. This is a different job from the logic ground on page 14 even though they are the same net; on a breadboard, route it separately and let them meet at the supply.

ON THE BOARD

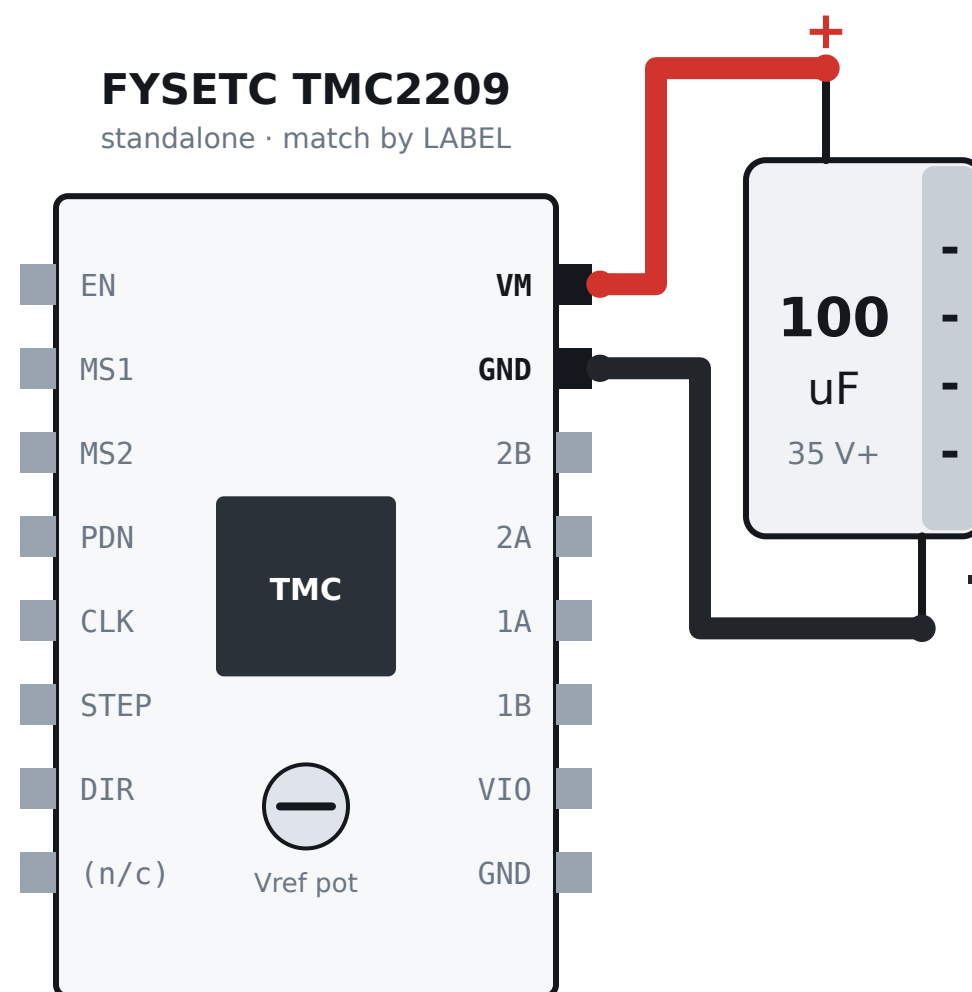
GND is row 27 column D. Bench PSU - on its own 22 AWG wire into row 27 column C.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

Bulk capacitor — connection 13

100 uF, at the driver's OWN pins

20 / 32



THE STRIPE IS THE NEGATIVE LEG.
Backwards, an electrolytic vents.

35 V minimum; your 50 V part is right. Never a 25 V cap on the 20 V rail — decelerating the drum pushes the rail up.

This capacitor supplies the sudden current the coils demand at each step, which the supply lead is too inductive to deliver. Legs as short as you can make them, bridging the driver's own VM and GND pins.

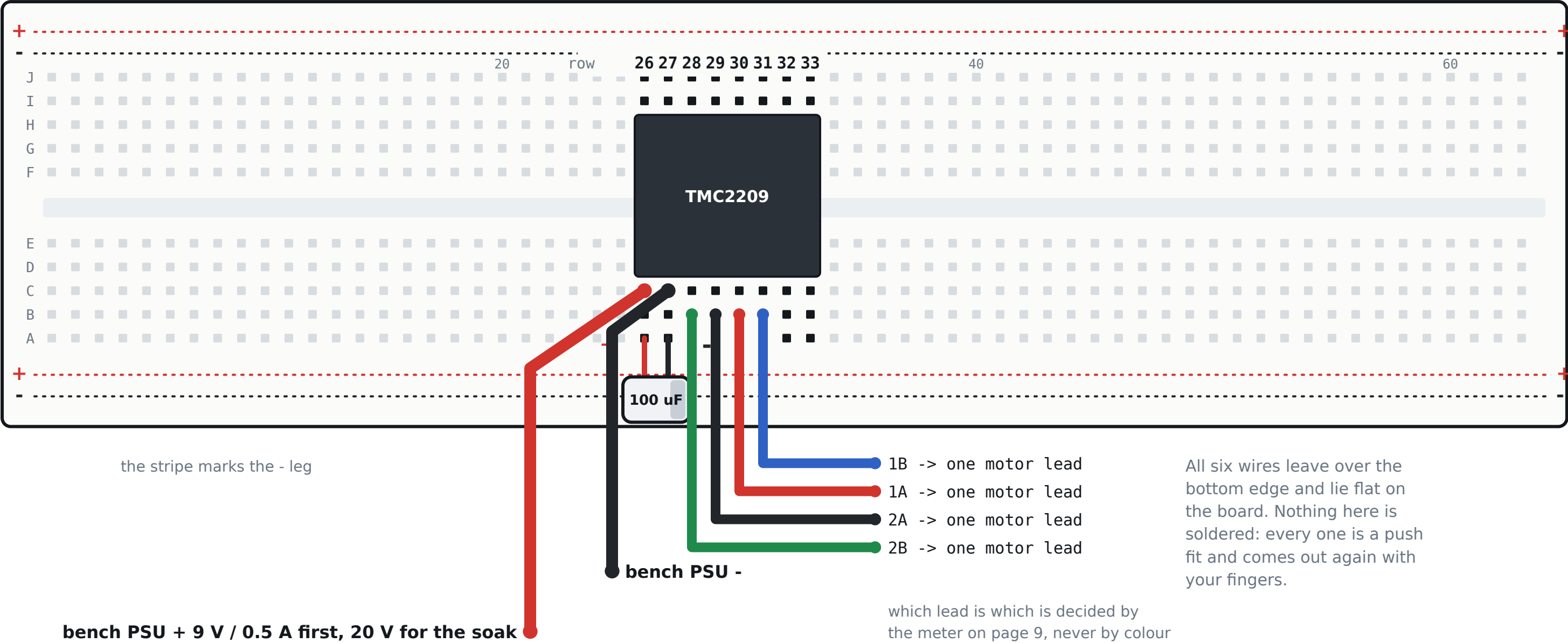
ON A BREADBOARD THIS MEANS THE TWO HOLES BESIDE THE MODULE, not the power rails at the end of the board. Twenty centimetres of breadboard wire defeats the capacitor entirely. This is the single thing breadboard builds get wrong.

Concretely, on this build: + leg row 26 column A - leg row 27 column A

drawn on page 22; lead prep on page 2

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

pin	pin is at	wire goes in	and reaches	
STEP	31 H	31 J	ESP GPIO8	M-F
DIR	32 H	32 J	ESP GPIO24	M-F
EN	26 H	26 J	ESP GPIO6	M-F
VI0	32 D	32 B	+ rail bot	M-M
GND (logic)	33 D	33 B	- rail bot	M-M
MS1	27 H	27 J	+ rail top	M-M
MS2	28 H	28 J	+ rail top	M-M
3V3 feed		a rail	+ rail top	M-F
GND feed		a rail	- rail top	M-F
+ rail link		far end	top <-> bot	M-M
- rail link		far end	top <-> bot	M-M
PDN, CLK	29, 30 J	NOTHING	stay empty	



Every power wire, by hole

what	pin is at	other end	wire goes in	
VM (driver)	26 D	PSU +	26 C	22 AWG
GND (power)	27 D	PSU -	27 C	22 AWG
100 uF +	-	bridges VM	26 A	legs 12 mm
100 uF -	-	bridges GND	27 A	stripe side
2B (coil)	28 D	pigtail lead	28 B	meter decides
2A (coil)	29 D	pigtail lead	29 B	meter decides
1A (coil)	30 D	pigtail lead	30 B	meter decides
1B (coil)	31 D	pigtail lead	31 B	meter decides

THE CAPACITOR IS NOT OPTIONAL AND ITS POSITION IS THE POINT. Rows 26 and 27 ARE the driver's VM and GND pins, which is why it belongs there and not on a rail twenty centimetres away.

0.7 A RMS needs none of this upgraded - page 2 says why. The coil current never travels along a rail: it runs in the four leads above and inside the driver. The supply wires are separate from the logic ground to keep the switching return out of it, not for current rating.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

THE CENTRE LEAD IS GND. That one is true whichever way you read the part, and it is the only thing on this page you cannot get backwards.

leads DOWN
branded face TOWARDS you



The terminal list, package UA

- 1 VCC supply, 3.0-24 V
- 2 GND the CENTRE lead
- 3 OUT open drain, sinks 25 mA

The datasheet numbers these in a view of the BRANDED face - the side carrying the supplier emblem, two digits of the part number and a temperature letter. Turn the part over and 1 and 3 change ends, which is the only way to get this wrong.

What it does, in its own words

- south pole turns the output ON
- output on = LOW (open drain, pulled down)
- north pole NOTHING AT ALL - not weaker, nothing
- B(OP) 95 G typical (50 min, 135 max)
- B(RP) 70 G typical - 25 G of hysteresis
- output open drain, 25 mA, leaks under 10 uA

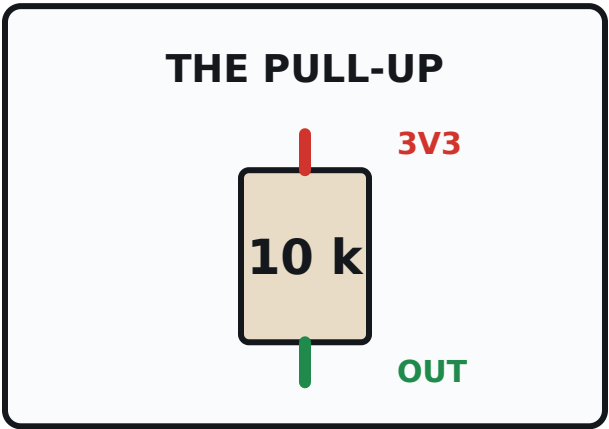
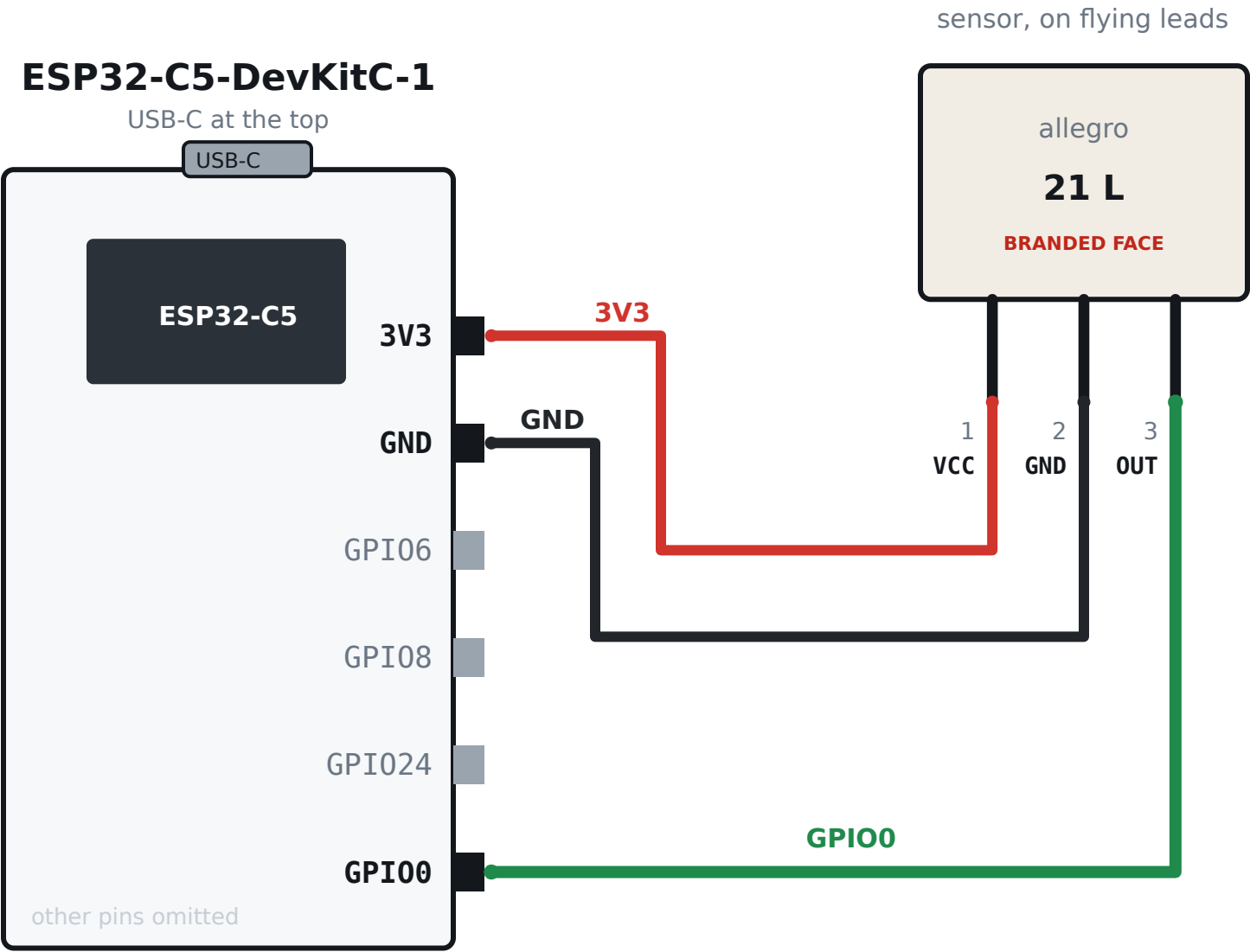
A NORTH POLE PRODUCES NO TRANSITION, so motion.hall_active_low cannot rescue a magnet glued in backwards - it inverts how the firmware reads an output that never moved.

Active LOW with the magnet present is now a DATASHEET fact

Spec 2 has carried VERIFY on this since the part changed from the A3144. The selection guide settles it: the column is headed "Output In South (Positive) Magnetic Field" and the A1121's entry reads "On (logic low)". So motion.hall_active_low = true, which is already the firmware default. What is still a bench question is the WIRING - a pull-up to the wrong rail, or a lead in the wrong hole, changes what the GPIO sees and not what the sensor does.

Wire the Hall sensor

Three leads and one resistor. VM may stay off for all of this.



THE OUTPUT IS OPEN DRAIN. It can pull the line down and it cannot pull it up, so with no resistor the GPIO floats and `hall` reads noise. This is the one wire in the guide that is a component rather than a jumper.

And 0.1 uF at the sensor

Across VCC and GND, at the sensor end of the flying leads. The datasheet: "strongly recommended ... in close proximity to the Hall element". The leads are long and the drivers are switching a metre away.

The four connections

VCC	->	3V3	M-F	3.0-24 V part; there is no 5 V rail
GND	->	GND	M-F	the same ground as everything else
OUT	->	GPIO0	M-F	column 0's hall pin, from hal/pins.h
10 k resistor	->	OUT to 3V3	-	the open drain cannot pull up by itself

THE SENSOR IS NOT ON THE BREADBOARD. It is on the module, on flying leads, so it has no row and column - which is also why this page carries no layout caveat: there is no layout to be wrong about.

ONE EVENING OF "THE SENSOR IS DEAD" IS ALWAYS A FLIPPED MAGNET. That is gotcha 2 in docs/ref/BOM.md and it has been there since the beginning. Check 2 below reads the glued magnet without touching it.

1 The sensor is alive, and which face is its SOUTH

Power the board. VM may be off. Type `hall`: with nothing near the sensor, column 0 reads raw=1 magnet=no. Now bring ONE FACE of a SPARE magnet up to the branded face. If it flips to magnet=YES, that face is the spare's SOUTH - mark it S with a pen, now. If not, turn the spare over; one of the two faces trips it.

2 The GLUED magnet's outward face, without prising it out

A magnet is its own polarity meter. Bring the spare's marked S face towards the glued magnet's exposed face. REPEL means the glued face is SOUTH and it will trip the sensor. ATTRACT means it is NORTH and it never will - the magnet is in backwards and has to be re-glued.

3 Turn the drum by hand and watch

`maint on` first, so nothing commands a move while your hands are on it. Turn the drum slowly through a full revolution with `hall` on screen. Once per revolution, and once only, column 0 must read magnet=YES.

CHECK 2, IN ONE TABLE

spare's S face and
the glued magnet...

REPEL	SOUTH - correct
ATTRACT	NORTH - re-glue it

Like poles repel. A spare whose south face pushes back is facing another south, and south is what turns this part on.

A wide arc of magnet=YES means the gap is too small or the magnet too strong for the bracket; the firmware does not care, but the edge is where a step count gets latched, so a narrow one is worth having. Record the gap.

air gap = _____ mm

Beep these five pairs BEFORE the first power-on

Meter on continuity (the beep symbol). Everything unpowered, USB out, motor still disconnected.
Five checks: three that must NOT beep, two that MUST. A mistake found here costs a minute; the same mistake found by powering up costs a driver.

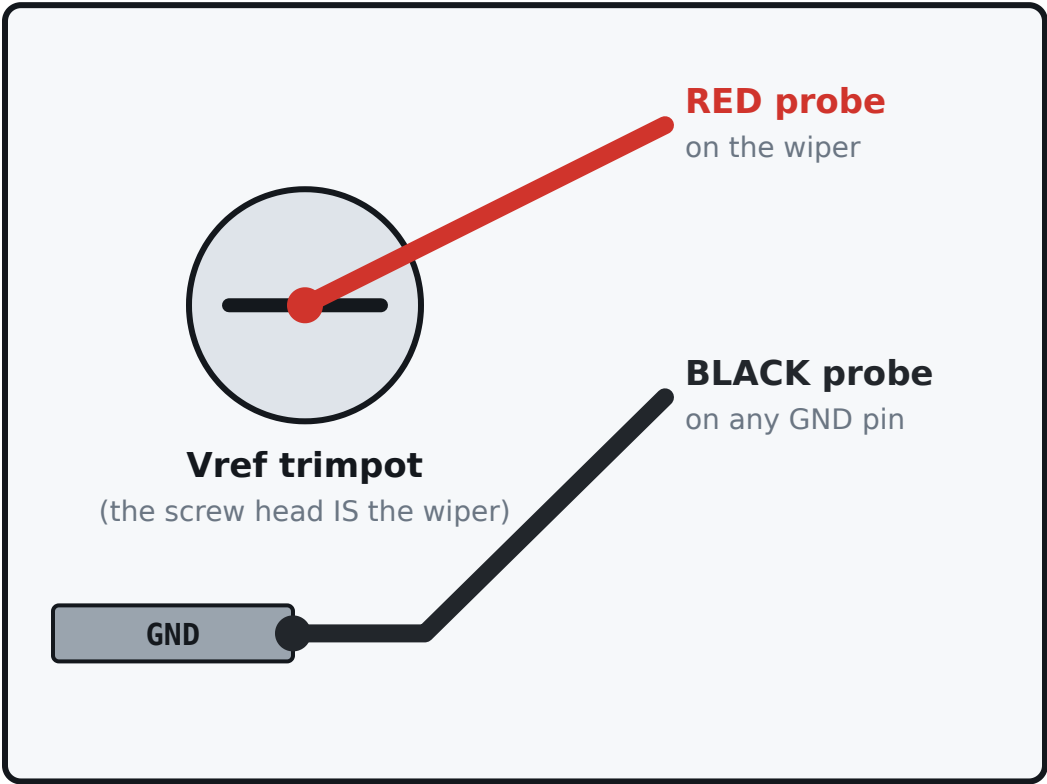
1	VM <--> GND MUST NOT BEEP	A short across the supply. This is the one that makes smoke.	☐
2	VM <--> 3V3 MUST NOT BEEP	Motor voltage on the ESP32's 3.3 V rail — destroys the board.	☐
3	3V3 <--> GND MUST NOT BEEP	Shorts the ESP32's regulator; the board will not boot.	☐
4	ESP GND <--> driver GND MUST BEEP	No common ground means every logic level is undefined. Nothing works, confusingly.	☐
5	ESP 3V3 <--> driver VIO MUST BEEP	MS1 and MS2 must beep to 3V3 as well — they are on the same net.	☐

All five correct? Then the first power-on cannot destroy anything through a wiring error. Go to page 27.

Motor disconnected. VM off. USB on. Driver NOT inserted.

THE DRIVER IS NOT IN THE BREADBOARD FOR THIS PAGE. It seats pot-down (page 7), so once it is in, the pot is against the board and unreachable. Set it lying on the bench on two flying leads, write the number down, and insert it already correct.

driver, close up



If your module has a labelled VREF pad, use that instead of the screw. Use a CLIP, not a hand-held probe: one slip from the wiper to a neighbouring pad kills the driver.

1. Read your sense resistor (page 6)

marking	R_sense	full scale	Vref for 0.7 A
R110	0.11 ohm	1.77 A	0.99 V
R150	0.15 ohm	1.35 A	1.29 V

- 2. VIO and GND on flying leads ONLY. USB in.
- 3. Meter to DC volts, 2 V range. Black on GND.
- 4. Turn the pot in small steps; set the figure above.

Vref set = _____ V

write it down NOW - it is unreadable once the module is seated

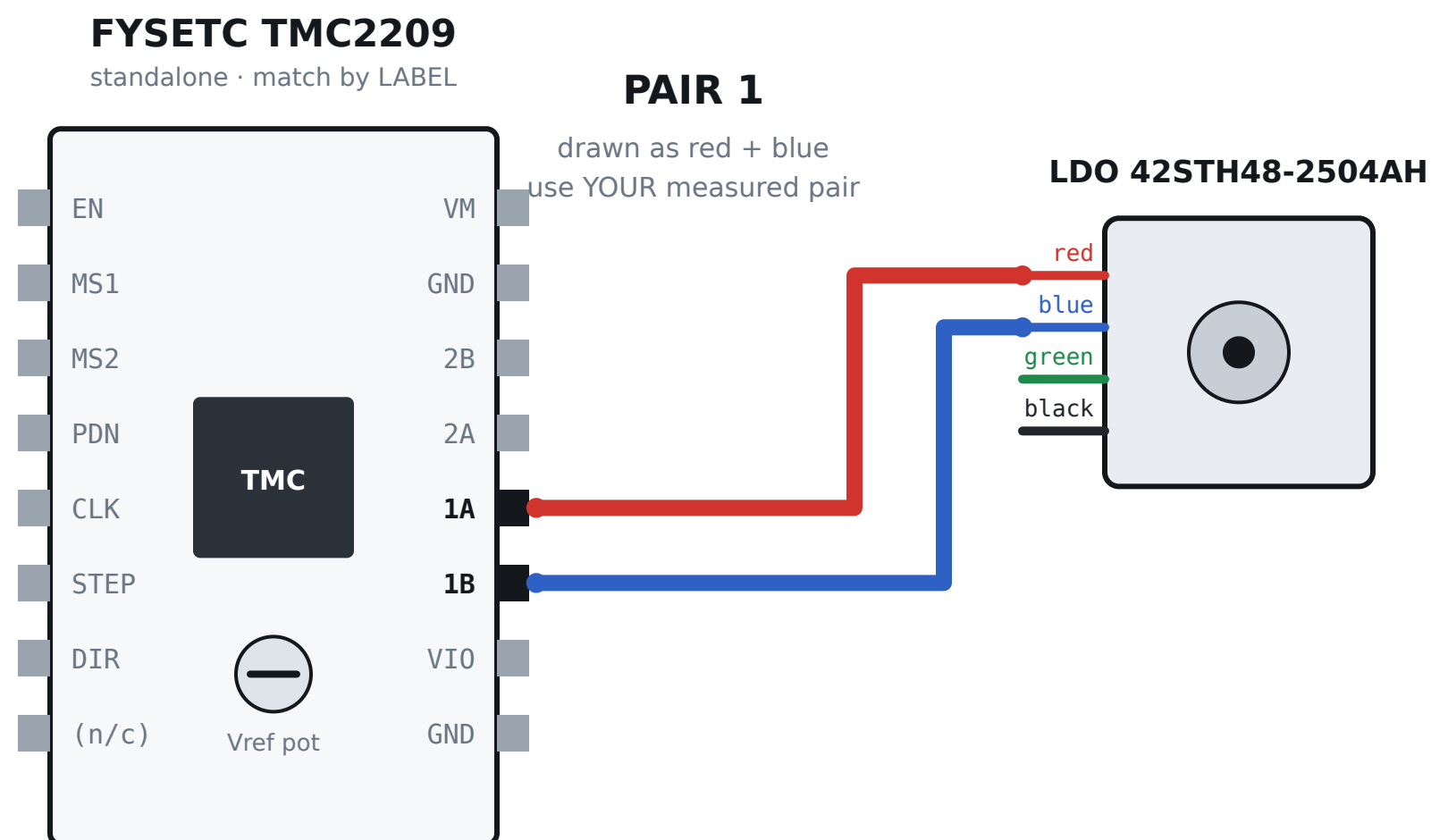
The pot is sensitive — a few degrees is 0.1 V. Creep up on the number; do not sweep past it and come back.

0.99 V is 0.7 A RMS on an R110 module. The A4988 formula would say 0.19 V and give you about 0.14 A — a motor that skips under load and looks too small for the job.

The TMC2209 uses VREF as a SCALING input against a full-scale current fixed by the sense resistor — it is not the A4988 formula, and using that one gives about 0.14 A, at which the motor skips under load and looks inadequate. The equation these numbers come from:

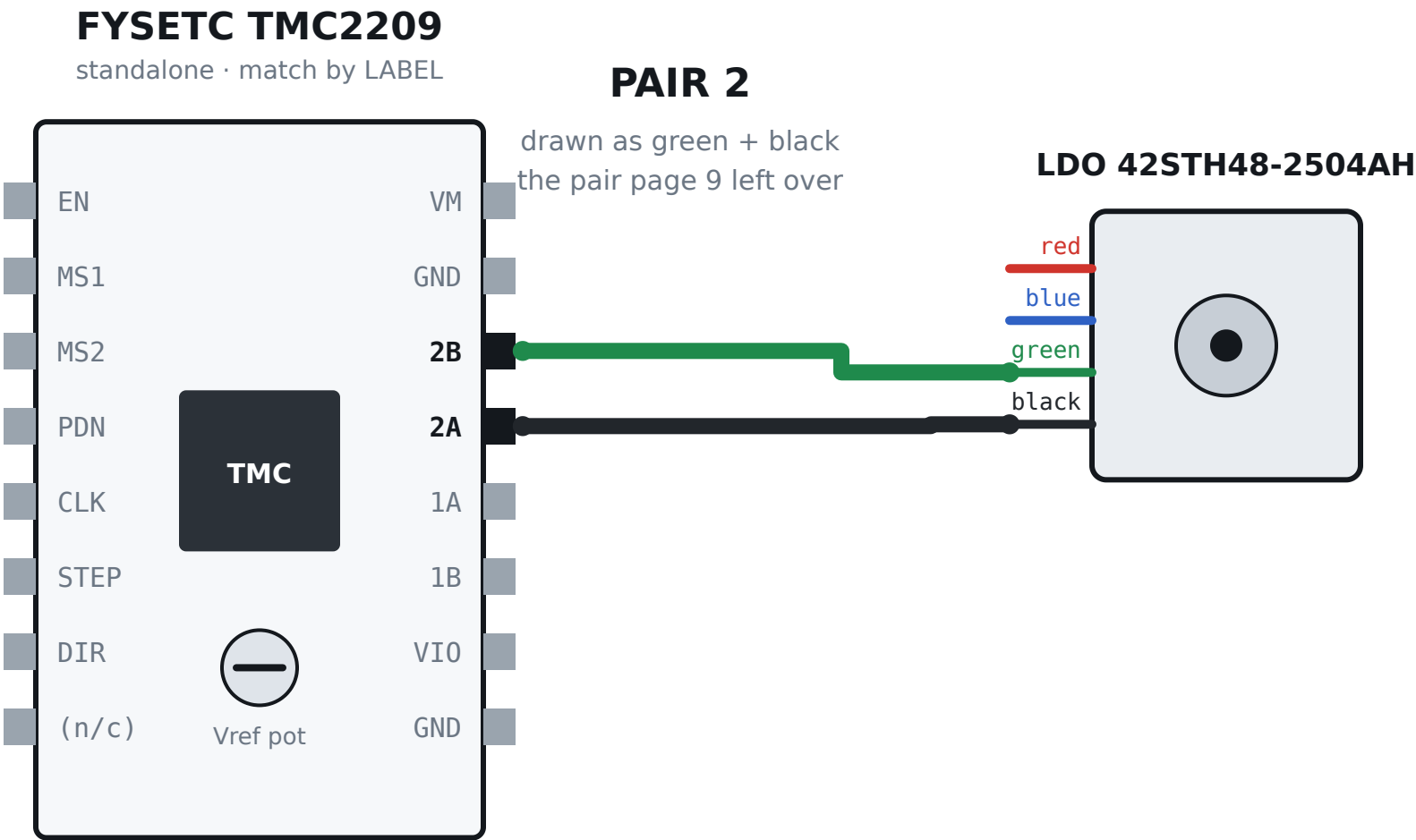
$$I_{RMS} = [0.325 / (R_{sense} + 0.02)] / \text{sqrt}(2) \times (Vref / 2.5)$$

MOTOR DISCONNECTED for this whole page. That is rule 3 on page 4: set the current before the motor can be subjected to it.



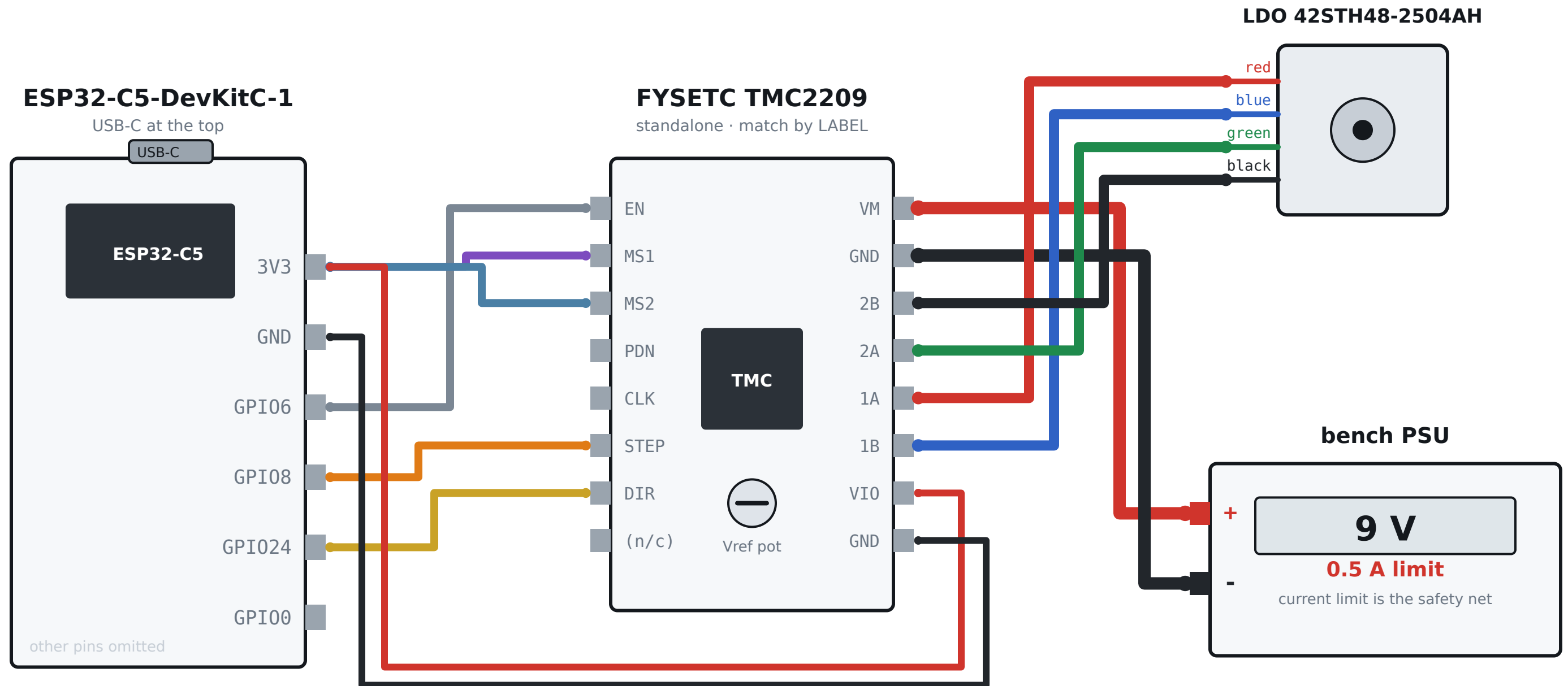
Both ends of ONE coil — the pair you found on page 9 — go to 1A and 1B. Which of the two ends goes to which pin does not matter; it only reverses rotation, and direction is a firmware bit. What matters is that this bridge gets both ends of one coil and never one end of each.

VM MUST BE OFF. Plugging a coil into a live driver is rule 1 on page 4, and it is the most common way these modules die.



The other coil, to 2A and 2B. Same rule: both ends of one coil into one bridge. Mixing them across the two bridges does not turn the motor — it buzzes, locks or judders, and it stresses the driver.

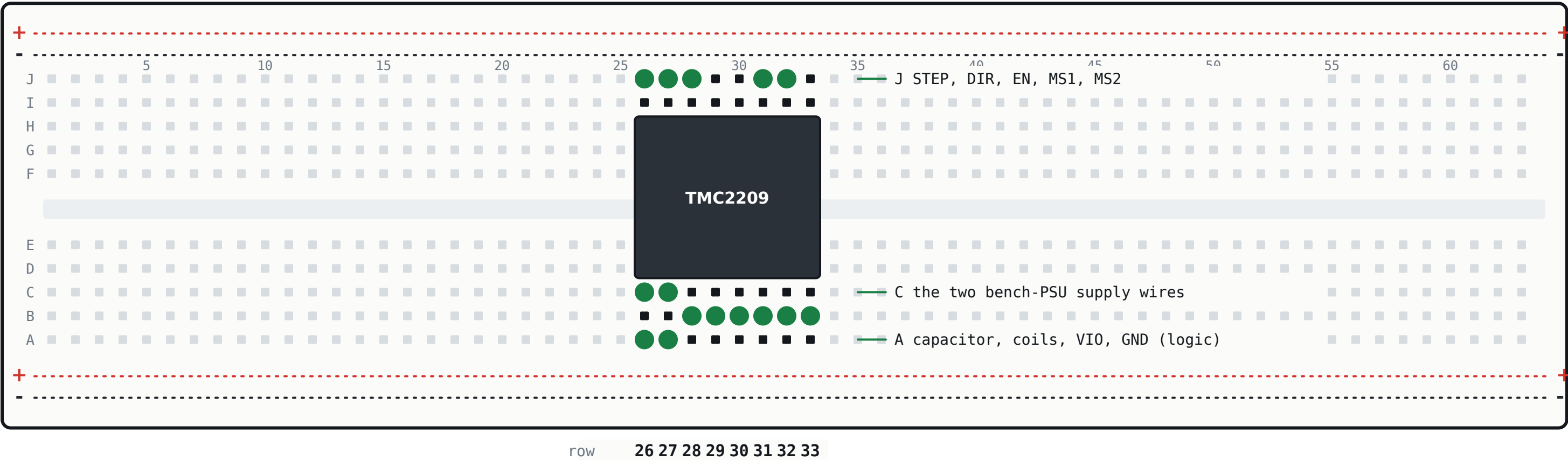
WITH THE MOTOR CONNECTED AND POWER STILL OFF, one last meter check: 1A to 1B beeps, 2A to 2B beeps, and 1A to 2A does NOT.



The geometry-proving check, once this is powered:

step 0 3200

must be EXACTLY one drum revolution. Two revolutions means MS1/MS2 are giving 1/8; four means 1/4. Judder with no net rotation means the coil pairing is wrong. Mark the drum first.



The whole board, row by row

☐	row 26	EN VM	A cap +, C VM in, J EN
☐	row 27	MS1 GND	A cap -, C GND in, J MS1
☐	row 28	MS2 2B	B 2B, J MS2
☐	row 29	PDN 2A	B 2A
☐	row 30	CLK 1A	B 1A
☐	row 31	STEP 1B	B 1B, J STEP
☐	row 32	DIR VIO	B VIO, J DIR
☐	row 33	(n/c) GND	B GND (logic)

Count before you power anything

- 11 jumper wires
- 2 supply wires from the PSU
- 2 capacitor legs
- 4 motor leads from the pigtail

19 things in the board, total

A hole you cannot account for is a wire in the wrong row. Pull it out and find it on page 21 or 22.

Then the meter, page 26. It finds what a count cannot: two wires in one row that should not share it.

POWER ON

- 1 Everything wired, MOTOR INCLUDED, nothing powered.
- 2 USB-C from the PC to the ESP32. VIO comes up; logic defined.
- 3 Confirm EN reads high (disabled) — page 12. USB in, VM still off.
- 4 Watch the console boot. Board healthy BEFORE any motor voltage.
- 5 maint on — nothing homes or moves while your hands are on it.
- 6 PSU to 9 V with a 0.5 A LIMIT set FIRST. Output stage now live.
- 7 en 1 — and only now are the coils energised.

STOP NOW — cut VM immediately

- any smell of hot plastic or varnish
- the driver too hot to keep a finger on
- the motor case climbing past uncomfortable early in a run
- the bulk capacitor warm, or bulging
- loud buzzing with no rotation (coil pairing — page 9)
- supply current spiking, or the supply cutting out
- the ESP32 rebooting repeatedly (brownout — check grounds)

Do not go looking for the console if something smells hot. Pull VM.

POWER OFF

- 1 en 0 — de-energise the coils first.
- 2 Remove VM.
- 3 Remove USB.
- 4 Before touching the motor plug: confirm the bulk cap has discharged.

THE BULK CAPACITOR HOLDS CHARGE after the supply is gone. That is still a live driver as far as rule 1 is concerned.

Standstill-current check (page 17)

With the motor energised and stationary, watch the supply current: it must STEP DOWN about a second after the last step. That is standstill reduction working, and it is the whole thermal contract. If it never steps down, PDN is tied the wrong way.

EXPECTED, not a problem: faults on columns 1-4 (nothing is wired to them), a steady hiss at standstill (the chopper), and the drum settling to a slightly different rest position after `en 0` — that is the 3.92 N·cm imbalance against a 2.2 N·cm detent, and seeing it confirms the premise. A no_hall fault on COLUMN 0 is NOT expected any more: this module has a sensor, so that is a finding and page 25 says which of three things it is.